TECHNICAL PROJECT REPORT

AutoDirector

Vertical Cinematic Feed & Storyboard Generator

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1. Executive Summary

In the contemporary digital landscape, short-form, vertical video content holds a dominant position across various entertainment platforms. However, pre-production workflows—specifically scriptwriting, shot-list formulation, and storyboarding—remain highly manual, labor-intensive, and time-consuming. **AutoDirector** is an end-to-end, multi-agent AI pipeline engineered to bridge this gap.

The system transforms a singular, raw descriptive sentence into a fully realized, structurally sound, 27-shot vertical cinematic storyboard in real time. By leveraging advanced Large Language Models (LLMs) and high-speed latent text-to-image diffusion models, AutoDirector automates creative division of labor via specialized autonomous agents, providing a scalable solution for content creators, independent filmmakers, and marketing professionals.

2. System Architecture & Multi-Agent Workflow

AutoDirector employs a decoupled multi-agent paradigm where individual architectural components function sequentially, using strict structured communication interfaces. The pipeline transitions from abstract contextual concepts to high-fidelity multimedia outputs.

1. Screenwriter	→	2. Cinematographer	→	3. Visual Artist	→	4. Streamlit UI
Gemini API converts an idea into a structured 3-Act narrative framework.		Gemini API builds 9 scenes & 27 granular vertical shot descriptors.		FLUX.1-schnell renders 27 customized high-res 9:16 storyboards.		Renders a dynamic, sequential, mobile-optimized vertical feed.

2.1 The Screenwriter Agent

The process begins with the ingest of a single-sentence user prompt via the interface. The **Screenwriter Agent**, powered by the Google Gemini API, assumes the persona of an expert Hollywood script consultant. It analyzes the raw seed idea and scales it into a classical 3-Act dramatic structure:

- **Act I (Setup / Exposition):** Establishes the core protagonist, world-building elements, and the inciting incident.
- **Act II (Confrontation / Rising Action):** Builds progressive complications, obstacles, and a narrative climax.
- **Act III (Resolution):** Handles the falling action, final resolution, and structural closure.

2.2 The Cinematographer Agent

Once the 3-Act structural script is established, the data is piped to the **Cinematographer Agent** (also powered by Google Gemini). This agent acts as a technical Director of Photography (DoP), parsing the text into a detailed shot list optimized explicitly for the 9:16 vertical video format. The structural output is deterministic:

- The narrative is divided into exactly **9 distinct chronological scenes** (3 scenes per Act).
- Each scene is broken down into exactly **3 compositionally varied camera shots**.
- The total yield is exactly **27 distinct camera shots**.

Each shot description generated includes detailed instructions on camera distance (e.g., Extreme Close-Up, Low-Angle Medium Shot), focal elements, lighting geometry (e.g., high-contrast chiaroscuro, cinematic neon rim lighting), and dynamic actor expressions to maximize vertical composition framing.

2.3 The Visual Artist Agent

The 27 visual descriptions serve as individual text prompt inputs for the **Visual Artist Agent**, which communicates asynchronously via the Hugging Face API with the **FLUX.1-schnell** model. FLUX.1-schnell is selected for its high performance, text-adherence, and fast generation speeds.

The agent appends persistent style-modifiers to each prompt to enforce aspect-ratio uniformity (9:16), cinematic consistency, photorealistic textures, and stylistic cohesion across all 27 frames, preventing stylistic drifting during generation.

3. Technology Stack Integration

Layer	Technology	Primary Responsibility
Frontend / UI	Streamlit Framework	Asynchronous rendering, sequential 9:16 display feed, UI state tracking.
Core LLM / Orchestrator	Google Gemini API	Structured text transformation, narrative construction, cinematic shot engineering.
Image Synthesis	Hugging Face (FLUX.1-schnell)	Real-time text-to-image synthesis, structural aspect adherence (9:16 rendering).
Runtime Environment	Python 3.x Engine	Asynchronous API thread execution, state management, orchestration.

4. Setup and Deployment Manual

Follow this technical guide to deploy the AutoDirector pipeline locally. Ensure that your environment has Python 3.8 or higher installed.

4.1 Repository Initialization & Installation

Clone the core repository to your local directory and install the pinned ecosystem dependencies using the pip package manager:

```
$ git clone https://github.com/your-repository/autodirector.git
$ cd autodirector
$ pip install -r requirements.txt
```

4.2 Environmental Configuration

The system requires external API authentication keys. Create a secure environment file named `.env` inside the root workspace folder and populate it with your active infrastructure credentials:

```
# AutoDirector Environment Configuration File
GEMINI_API_KEY=your_gemini_key_here
HUGGINGFACE_API_KEY=your_huggingface_key_here
```

4.3 System Execution

Initialize the local web application server using the Streamlit execution layer. The application automatically builds and displays the interface on your local host loopback:

```
$ streamlit run app.py
```

> **Operational Notice**
>
> Upon launching, access the UI container through your browser at `http://localhost:8501`. Enter your text idea into the input terminal to initialize the generative multi-agent workflow.

5. Conclusion and Future Work

AutoDirector presents a highly structured approach to automated pre-production asset generation. By organizing large language engines and image synthesis networks into a specialized, multi-agent format, the system cuts down the pre-production timeline from hours to minutes.

Future system updates aim to focus on introducing persistent seed-control techniques to maintain character consistency across all 27 frames, adding automated localized script dialogue overlays, and integrating synthetic temporal video-to-video generation blocks to convert the static vertical storyboard into an assembled cinematic rough-cut trailer.